Review Article - Narrative Review

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Optic disc and peripapillary tumors

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Abstract:

This review covers the tumors either originating or seen close to the optic disc in the peripapillary area. Most of the optic disc tumors are diagnosed on clinical examination. In a few atypical cases, ancillary imaging, such as optical coherence tomography, fluorescein angiography, and ultrasonography, may be required to demonstrate characteristic features. The majority of these tumors have systemic associations, and hence, a multidisciplinary approach involving ocular oncologists, neurophysicians, radiation oncologists, and medical oncologists is the need of the hour. The treatment of optic disc tumor is challenging due to its proximity to critical areas such as optic nerve and macula, which could be a limiting step when compared to conventional treatments such as laser photocoagulation and surgical excision.

Keywords:

Juxtapapillary, optic disc, peripapillary, retinal capillary hemangioblastoma, retinoblastoma, tumors

Introduction

Numerous lesions can cause elevation of the optic disc. Optic disc tumors are rare benign or malignant lesions that primarily affect vital vision structures, such as the optic nerve and macula, and may be associated with life-threatening systemic associations. Hence, early diagnosis and timely intervention are important in the management of optic disc tumors. Tumors involving the optic nerve head may be primary, contiguous with adjacent structures such as the retina and choroid, or metastatic lesions from distant sites.

Methods

A literature search was performed for the English language articles using PubMed, Scopus, and Google Scholar, for any date up to December 2023. The following keywords were used "optic disc tumors," "peripapillary tumors," "Juxtapapillary tumors," "retinal tumors," and "choroidal tumors." Search results were initially reviewed by the title and abstract, and the articles were selected for more in-depth analysis if deemed relevant. Pertinent articles were examined in depth for this report. Additional articles were retrieved from bibliography of the articles searched by the above keywords. Inclusion and exclusion of the article in the text were based on relevance.

Classification of Tumors of the Optic Disc according to Cell of Origin

- Neurosensory retina: Retinoblastoma (Rb) and medulloepithelioma
- Vascular tumors: Capillary hemangioblastoma, cavernous hemangioma, racemose hemangioma, and choroidal hemangioma
- Melanocytic tumors: Choroidal melanoma, optic disc melanocytoma, combined hamartoma of retina and retinal pigment epithelium (RPE), adenoma and adenocarcinoma of RPE
- Glial tumors: Astrocytic hamartoma, acquired retinal astrocytoma
- Lymphoid origin: Lymphoma, leukemia
- Metastasis: From primaries (lung, breast, etc.)

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- Anterior extension of retrobulbar optic nerve tumors: Juvenile pilocytic astrocytoma, optic nerve sheath meningioma.

Retinoblastoma

Rb is the most common intraocular tumor in children, accounting for approximately 11% of cancers occurring in the 1st year of life, with 95% diagnosed before 5 years of age. Rb is a genetic disease; inactivation of both alleles of Rb susceptibility gene (*RB1*) predisposes an individual to the disease.[1]

Clinical features

Rb can spread to involve the optic disc. It can also occur in the parapapillary area [Figure 1a], or a large exophytic tumor may obscure the optic disc involvement during fundus examination. Obscuration of the optic nerve during diagnosis may indicate postlaminar optic nerve invasion. During treatment, persistent and complete obscuration of the optic nerve due to Rb is generally considered to have a poor prognosis for both vision and globe salvage.[2] Clinical features such as the presence of buphthalmos, neovascularization of the iris, and secondary glaucoma can also indicate (pre/postlaminar) optic nerve invasion.[3,4]

Imaging

Imaging modalities are essential for identifying the key features that help diagnose Rb and determine the appropriate treatment.

Ultrasonography

Ultrasonography (USG) is often helpful in diagnosing Rb, generally appearing more homogeneous associated with intraocular seeding and the presence of multiple hyperechoic lesions within the tumor suggestive of tumor calcification [Figure 1b].

Orbital imaging

Magnetic resonance imaging (MRI) of the orbit is preferred over computed tomography (CT) because of less radiation risks and its ability to differentiate Rb from other simulating conditions.[5] In addition, MRI has a higher resolution and provides greater tumor details, including ruling out optic nerve and central nervous system (CNS) involvement and associated pineal gland tumors. Imaging may reveal an orbital mass contiguous with the eyeball or a thickened optic nerve [Figure 1c]. It is crucial to compare the optic nerve size in the other eye in axial, coronal, and sagittal sections before commenting on optic nerve infiltration. Subtle optic nerve extension may be seen as thickening of the base of the optic nerve. A posterior staphyloma evident on imaging or loss of posterior rounded contour of the eyeball are highly suspicious of extraocular extension.[6]

Management

The management of Rb involving the optic disc is complex and is based on clinical circumstances. Peripapillary Rb or overhanging tumors obscuring the optic disc will require brain MRI with cerebrospinal fluid (CSF) analysis to rule out metastasis. Patients must undergo baseline systemic evaluation to rule out metastasis. Treatment is determined by optic nerve involvement on MRI. Enucleation is the primary treatment for optic disc involvement and is often supplemented by chemotherapy or irradiation based on histopathology. If the extension is beyond the optic disc, it is preferable to administer neoadjuvant chemotherapy followed by enucleation and adjuvant chemotherapy and

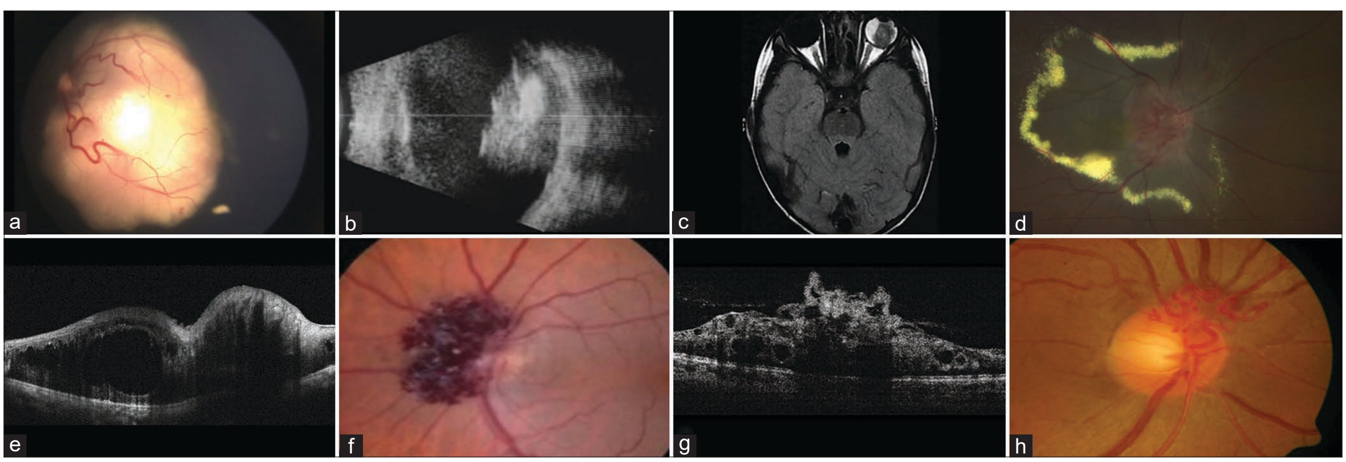

Figure 1: Peripapillary retinoblastoma (Rb) and retinal vascular tumors. (a) Fundus photograph of the left eye showing a large, yellowish lesion at the posterior pole abutting the optic nerve with dilated tortuous feeder vessels characteristic of Rb. (b) Ultrasonography showing a solid, echogenic intraocular mass with variable shadowing due to calcification and heterogeneity due to necrosis and hemorrhage in a case of Rb. (c) Magnetic resonance imaging of the orbit revealing an exophytic intraocular Rb adjacent to optic nerve but not infiltrating it. (d) Fundus photograph showing an orange-colored lesion of Juxtapapillary retinal capillary hemangioma with a ring of subretinal exudation. (e) Optical coherence tomography (OCT) showing intraretinal fluid with cystoid changes in the peripapillary area causing posterior shadowing. (f) Fundus photograph showing a grape-like cluster of Juxtapapillary retinal cavernous hemangioma. (g) OCT line scan through the lesion showing disorganized retinal layers with numerous cystic spaces separated with septa. (h) Fundus photograph of the right eye showing a dilated and tortuous vessel over the optic disc characteristic of racemose angioma

radiotherapy. Prelaminar/laminar involvement of the optic nerve has a good prognosis, whereas the prognosis is relatively worse with postlaminar involvement and far worse when the tumor extends to the optic nerve transection.[4,7]

Chemotherapy

Neoadjuvant intravenous chemotherapy is required if extensive optic nerve involvement is observed on MRI orbits. This should be followed by enucleation and adjuvant chemotherapy. If optic nerve infiltration is detected after enucleation, adjuvant intravenous chemotherapy is beneficial in minimizing the risk of metastasis. With the advent of prophylactic adjuvant intravenous chemotherapy, the survival rate of patients with optic nerve tumor infiltration has improved compared with that in the prechemotherapy era.

Radiotherapy

With the advent of systemic chemotherapy, the role of radiotherapy as a primary treatment for Rb has become limited. In cases with orbital involvement, adjuvant external beam radiotherapy (EBRT) along with 12 cycles of high-dose intravenous chemotherapy is recommended.[6,8] Newer image-guided intensity-modulated radiotherapy can deliver a therapeutic dose to the tumors, minimizing adjacent tissue damage compared with EBRT.[9,10] The role of proton-based radiotherapy has also been explored for treating orbital Rb with a high local tumor control rate.[11]

Medulloepithelioma

Medulloepithelioma, also known as diktyoma, is a tumor that originates from the neuroepithelium of the developing ciliary body epithelium in adults. Rarely, it can involve the posterior most part of the optic nerve.[12,13] The tumor usually occurs in children aged 2–4 years; however, there is a lag in the diagnosis because of the rarity of the tumor and the onset of symptoms.[14]

Clinical features

Clinically, it is seen as a yellow–white rounded lesion. Optic nerve involvement of medulloblastoma can occur at the disc, which may closely resemble astrocytic hamartoma and Rb. It can present with proptosis whenever there is extensive involvement of the optic nerve.[12] Although some are malignant, distant metastases are rare. The mortality rate is up to 10%, and death is usually caused by intracranial extension from secondary local tumor recurrence.[12,15]

Management

Enucleation is the treatment of choice for medulloepithelioma. A definitive diagnosis may often be established on the basis of histopathological findings. Optic nerve medulloepithelioma that is inaccessible to biopsy or local resection requires enucleation.[12,16] Incisional biopsy and local resection can be considered whenever a nonneoplastic differential diagnosis exists. Radiotherapy can be considered for localized tumors and enucleation for extensive tumors.[14,16]

Juxtapapillary retinal capillary hemangioblastoma

Juxtapapillary retinal capillary hemangioblastoma (JRCH) is an uncommon benign vascular hamartoma. They can manifest either sporadically or as a part of von Hippel–Lindau (VHL) syndrome.[17] JRCHs are small nodular vascular hamartomas seen over the optic disc or located in the Juxtapapillary region [Figure 1d]. Isolated retinal capillary hemangioblastoma (RCH) is usually stable and unassociated with vision-threatening complications and hence can be safely observed.[18] Compared with peripheral RCH, JRCH is less commonly seen; however, because of their location, JRCH is easily misdiagnosed before they exhibit endophytic growth protruding into the vitreous cavity, resulting in visual impairment secondary to macular exudation, subretinal fluid (SRF) accumulation, macular edema, epiretinal membrane formation, and exudative or tractional retinal detachment.[19]

Genetic importance

VHL gene is a tumor suppressor gene and patients with VHL have an autosomal dominant inheritance. Point mutation alpha domain of VHL gene is responsible for JRCH (missense mutation).[20,21] VHL mutation causes an increase in hypoxia-inducible factor (HIF-α). HIF-α increases the expression of proangiogenic factors, which have a role in tumor development.

Clinical features

Most JRCHs are asymptomatic and remain undetected. These lesions can be indistinct on clinical examination, sometimes manifesting with only a subtle bulge at the disc margin, which often affects the temporal side of the optic disc and is often misdiagnosed as disc edema. Compared with peripheral RCH, there is no feeding artery and draining vein which could delay diagnosis.

Spectral-domain optical coherence tomography

Spectral-domain optical coherence tomography (SD-OCT) is a noninvasive method for tumor assessment. JRCH can cause posterior shadowing due to vascularity and higher flow. OCT can demonstrate photoreceptor layer integrity and the extent of the epiretinal membrane and intraretinal and SRF associated with JRCH. Exophytic JRCH is typically observed in intraretinal layers, predominantly outer retinal layers that grow into the subretinal space [Figure 1e], whereas endophytic JRCH typically extends from the inner retinal surface into the vitreous.[22] It is also a valuable tool for differentiation from other simulating lesions such as astrocytic hamartoma and combined hamartoma of the retina, which usually have a gradual smooth transition, unlike RCH.[23,24] Similarly, OCT-angiography (OCTA) can assess the intrinsic vascularity of a tumor and help differentiate it from other vascular tumors.[25] The presence of middle/outer retinal vascularity on OCTA effectively excludes the diagnosis of peripapillary choroidal neovascular membrane (CNVM).

Fundus fluorescein angiography

JRCH lesions generally show an early hyperfluorescence and progressive late leakage on fundus fluorescein angiography (FA). The individual vessels of exophytic JRCH can be distinguished in the arteriovenous (AV) FA phase but are obscured by leakage in the late phase. In contrast, endophytic JRCH profusely leaks fluorescein, even in early frames. Notably, disc edema can be associated with the engorgement of disc vessels, which can be prominent on angiography; however, these engorged disc vessels are typically confined to the disc borders and often encompass the entire optic disc.[22] It can also help identify subclinical lesions in the periphery and areas of peripheral nonperfusion and thus predict neovascularization in future.[26,27]

Treatment

Asymptomatic patients are usually observed because the location of JRCH makes treatment very difficult. Asymptomatic small lesions (up to 500 μm) not associated with exudation or neurosensory detachment require close observation. The clinical course of JRCH is usually unpredictable and can progress rapidly. However, spontaneous regression of JRCH has also been observed.[28] Patients with symptoms require treatment based on the size, extent, presence of exudation, and associated VHL disease.

The treatment of choice for JRCH is photodynamic therapy (PDT), which causes fibrosis and involution of the smaller JRCH.[29,30] Due to the recent unavailability of verteporfin dye, indocyanine green-enhanced transpupillary thermotherapy (TTT) is a viable option for PDT causing tumor fibrosis and resolution of SRF.[31,32] Intravitreal antivascular endothelial growth factor (VEGF) injections, such as bevacizumab and ranibizumab, when used in combination with PDT or TTT, inhibit the growth of RCH lesions by suppressing VEGF, thereby reducing the exudation and preserving the structural integrity of retinal layers.[33,34] Similarly, the antiangiogenic properties of steroids have also shown benefits when combined with PDT or TTT.[35]

Systemic therapy with drugs such as sunitinib has been attempted in patients with VHL-associated RCH, but with no real benefit in suppressing the tumor growth.[36] Recently, use of belzutifan, an oral HIF-2α inhibitor, has been tried to treat VHL-associated CNS hemangioblastoma and RCH.[37,38] JRCH associated with macular pathologies such as epiretinal membrane, tractional retinal detachment secondary to glial membrane formation causing macular traction, or, more frequently, total exudative detachment may require vitreoretinal surgery.[39,40] Refractory symptomatic cases with total exudative retinal detachment can be treated with EBRT to decrease the tumor volume and stabilize retinal detachment.[41] Rarely, microsurgical ligation of the JRCH feeder vessel is another treatment option where PDT and anti-VEGF fail to cause tumor regression.[42]

Screening and follow-up

Detection of JRCH can be the first manifestation of VHL disease. JRCH should prompt dilated fundus examination and widefield FA every year in all patients. Family screening of all the members should be performed at baseline in all patients diagnosed with JRCH to diagnose underlying VHL disease.[43]

Juxtapapillary retinal cavernous hemangioma

Retinal cavernous hemangioma is a rare, nonprogressive vascular hamartoma. It can occur as an isolated, solitary lesion or as a component of an autosomal dominantly inherited oculoneurocutaneous syndrome.[44,45] The tumors are more commonly seen in young women (mean age, 23 years) with unilateral presentation in 90% of cases. In most cases, the lesion remains stable with rare tumor growth and progression associated with vitreous hemorrhage seen in few patients.[15,44,46]

Clinical features

A cavernous hemangioma of the optic disc is a clinical diagnosis. Typical fundus findings are clumps of reddish–blue saccular aneurysms (cluster of grapes appearance) with fibroglial tissue over the tumor, absence of feeder vessels, and lack of exudation [Figure 1f]. Macular involvement and vitreous hemorrhage are rare and are usually precipitated by strenuous work or during labor.[47-49] Contraction of the fibroglial component of cavernous hemangioma can cause transient episodes of vitreous hemorrhage with secondary fibrosis. Visual symptoms are more often associated with cavernous hemangioma involving the posterior pole.

Optical coherence tomography

OCT of a retinal cavernous hemangioma shows a disorganized epiretinal membrane with cystic spaces within the inner and outer retinal layers, representing saccular aneurysms [Figure 1g].[47,50,51] OCT is useful for assessing the progression of these lesions. OCT findings of a cavernous hemangioma of the optic disc show numerous interconnected, blood-filled vascular saccules within inner retinal layers, separated by thin septa.[52,53] Similarly, OCTA can noninvasively identify cavernous hemangioma without fluorescein dye and provide dynamic lesion information showing a characteristic reverse fluorescein cap sign.[50,51,54]

Fluorescein angiography

On FA, cavernous hemangioma is observed during the mid- and late venous phases without evidence of leakage. A characteristic plasma–erythrocyte level occurs because the sedimentation of erythrocytes in the dependent portions of the aneurysms blocks fluorescence, whereas the plasma in the superior portion of the lesion fluoresces intensely, showing a characteristic fluorescein cap sign.[55] Histology of cavernous hemangiomas have large endothelial-lined vascular channels with thin walls and an intact blood–retinal barrier; hence, no leakage is observed in the late phase of angiogram.[56]

Management

Optic disc cavernous hemangioma is a very slow-growing tumor. Most lesions are asymptomatic; hence, observation with yearly follow-up is required. Association of optic disc cavernous hemangioma with a history of seizure will warrant brain MRI to look for associated intracranial hemangioma. Nonresolving vitreous hemorrhage may require vitrectomy, and the peripapillary portion of the tumor can be excised to relieve vitreopapillary traction.[46,57]

Retinal racemose hemangioma

Congenital AV malformation or racemose hemangioma of the optic disc (angioma racemosum) is a rare, unilateral, and sporadic condition characterized by the appearance of dilated and tortuous retinal vessels commonly extending from the optic disc to the retinal periphery [Figure 1h].[58] There is no hereditary tendency. It can be a part of Wyburn–Mason syndrome, which is characterized by similar vascular malformations in the midbrain, maxilla, mandible, and pterygoid fossa.[15]

Clinical features

Three different groups of AV malformations have been described by Archer *et al*., each with unique characteristics.[59] The first group is typically asymptomatic and is characterized by an abnormal capillary plexus between the major blood vessels. The second group, which lacks any intervening capillary bed between the artery and vein, is associated with retinal complications and a low risk of intracranial AV malformations. Finally, the third group which is characterized by extensive AV malformations with dilated and tortuous vessels, with no distinction between the artery and vein. Understanding these groups' differences is important to diagnose the disease early and screen for systemic associations. Most cases are asymptomatic. The cause of decreased vision is ocular complications such as vascular occlusions, vitreous hemorrhage, and rare congenital disc anomalies.[60-62]

Imaging

OCT scan through the disc revealed multiple large intraretinal vessels with back shadowing with/without associated macular edema.[63] OCTA of the disc shows dilated and tortuous vessels in the superficial retinal layers with increased intrinsic intraretinal vascular overlay.[62,64]

Treatment

Most of the cases are asymptomatic; hence, observation is warranted. Neuroimaging to detect any concurrent orbital and cerebral vascular malformations is advised for patients with retinal AV malformations.

Choroidal hemangioma-circumscribed and diffuse variant

Choroidal hemangioma is a benign vascular hamartoma commonly involving the optic disc.[65] Circumscribed choroidal hemangioma (CCH) is a unilateral, orangish, solitary tumor located at the posterior pole [Figure 2d], whereas diffuse choroidal hemangioma (DCH) presents as a unilateral tumor with exudative retinal detachment and is associated with Sturge–Weber syndrome, which is characterized by vascular malformations involving the skin, brain, and eye [Figure 2g].[66]

Clinical features

Both circumscribed and diffuse variants can involve the optic disc. Patients usually present with decreased vision depending on the size and location of the tumor. CCHs are usually located temporal to the optic disc. In most cases, they remain asymptomatic; however, in cases the tumor is seen beneath the fovea or the SRF is seen within the macula, patients might complain of distorted and blurred vision [Figure 2e].[67] Associated clinical features include exudative retinal detachment, macular edema, pigmentary changes within the RPE, subretinal fibrosis over the lesion's surface, and orange pigment.[68]

The thickened choroid in DCH around the disc can cause disc cupping. DCH is associated with developmental anomalies of the angle and increased episcleral pressure, which can lead to increased intraocular pressure.[69] Exudative retinal detachment is an eventual complication in many eyes with DCH, which if left untreated can lead to total retinal detachment and secondary neovascular glaucoma.[70]

Ultrasonography

USG is an excellent tool for differentiating Juxtapapillary choroidal hemangiomas from their mimics, such as choroidal metastases and amelanotic choroidal melanoma. It shows characteristic high internal reflectivity within the tumor because of its heterogeneous nature [Figure 2f and h].[71]

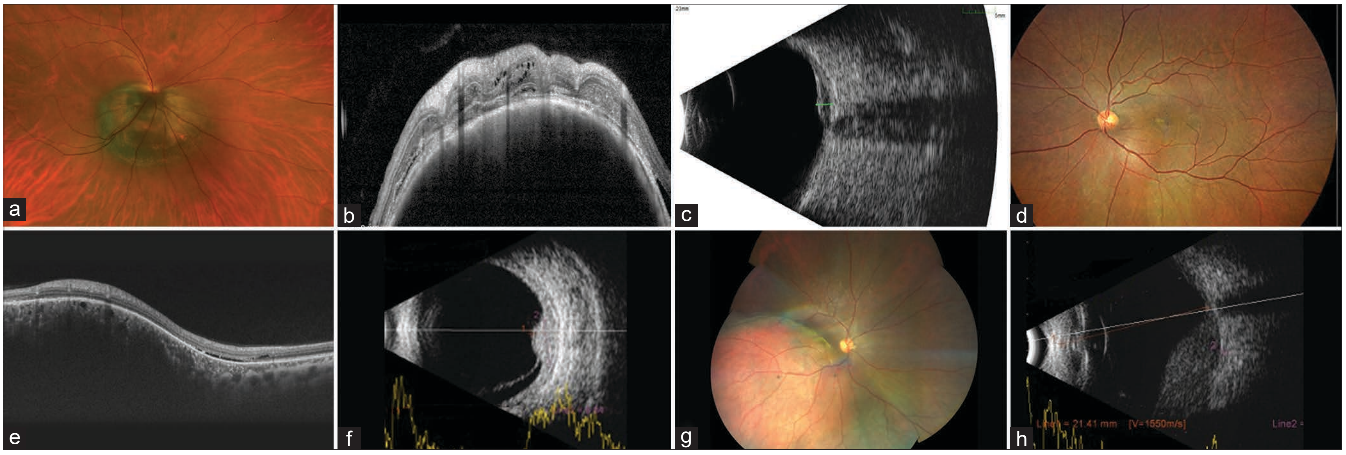

Figure 2: Peripapillary choroidal melanoma and hemangioma. (a) Fundus photograph of the right eye showing a large circumpapillary pigmented choroidal melanoma. (b) Optical coherence tomography (OCT) showing a dome-shaped elevated lesion with compression of the overlying choriocapillaris along with intraretinal and subretinal fluid (SRF), in a case of peripapillary choroidal melanoma. (c) Ultrasonography (USG) of an eye with choroidal melanoma showing an elevated choroidal lesion with impingement on the adjacent optic nerve head. (d) Fundus photograph of the left eye showing a solitary, circumscribed orangish tumor with surrounding SRF, suggestive of choroidal hemangioma. (e) OCT showing an elevated, smooth dome-shaped lesion with enlarged choroidal vessels, minimal sub- and intraretinal fluid, in a case of choroidal hemangioma. (f) USG showing an elevated choroidal lesion with high internal reflectivity and surrounding exudative retinal detachment suggestive of choroidal hemangioma. (g) Fundus photograph of the right eye in a patient with Sturge-Weber syndrome showing a diffuse orangish choroidal lesion adjacent to the optic disc suggestive of diffuse choroidal hemangioma. (h) USG showing diffuse choroidal thickening with a solid, echodense choroidal lesion adjacent to the optic nerve head suggestive of peripapillary nodular component in a case of diffuse choroidal hemangioma

Treatment

Observation is typically considered when the Juxtapapillary choroidal hemangiomas are small, asymptomatic, and not causing significant vision problems. TTT can be effective in treating small CCH with shallow SRF, with a tumor base <10 mm and tumor thickness <2.5 mm.[65] PDT is the preferred treatment for CCH with more than 95% success in local tumor control and resolution of SRF.[68] Oral propranolol is used in a few selected cases of DCH with exudative retinal detachment with limited success.[72,73]

EBRT is the treatment of choice for large Juxtapapillary DCH with exudative retinal detachment.[74,75] Plaque brachytherapy is usually reserved for Juxtapapillary tumors more than 2.5 mm height not amenable for PDT.[76] Proton beam radiotherapy has also been shown to be an effective option in treating exudative RD and decreasing the tumor volume.[77]

Juxtapapillary choroidal melanoma

Uveal melanoma can extend from neighboring structures to the optic disc. Optic nerve invasion is rare and occurs in only 0.6%–7% of cases.[15]

Clinical features

Clinical presentation varies with tumor size. It usually presents as a grayish-brown elevated lesion in the peripapillary area [Figure 2a] with compression of overlying choriocapillaris as seen on OCT [Figure 2b]. Amelanotic melanomas account for 20%–25% of all melanomas and are thought to arise from an amelanotic nevus.[78] Due to its lack of pigmentation on clinical examination, it can be easily misdiagnosed and hence multimodal imaging helps to confirm the diagnosis. Patients with associated macula involvement or exudative RD can present with decreased visual acuity. The clinical finding of an overhanging tumor obscuring at least a portion of the optic disc is a unique feature of Juxtapapillary melanomas.[79] The two important factors correlating with postlaminar optic nerve invasion are poor visual acuity and increased intraocular pressure, which require further imaging.[80-82]

Ultrasonography

USG, the primary imaging technique for choroidal melanoma, aids in the diagnosis and determination of tumor characteristics. Tumor dimensions, such as apical height and basal diameter, are essential for treatment planning. B scan aids in determining the local extrascleral extension and optic nerve involvement of the Juxtapapillary melanoma [Figure 2c].

Magnetic resonance imaging

MRI can be used to assess the extent of Juxtapapillary melanoma and its relation to surrounding structures, especially to rule out extraocular extension.

Treatment

Juxtapapillary melanoma presents a therapeutic challenge in preserving vision, preventing complications, and controlling tumors. Enucleation is a common treatment protocol for peripapillary tumors >180° or more of optic disc margin involvement or large overhanging choroidal melanoma.[81]

Radiation therapy is a globe-sparing alternative to enucleation in the case of Juxtapapillary melanoma.[83,84] Plaque radiotherapy with ruthenium-106, palladium-103, or iodine-125 for Juxtapapillary melanoma provides local tumor control in approximately 80% of eyes at 10 years.[79,85,86] The customized design of a notched and slotted plaque instead of a round plaque achieves an effective radiation field to tumor avoiding optic disc.[79,87] Notched radioactive plaques showed less recurrence (6%) as compared than round plaques (38%).[79]

EBRT offers better tumor control than brachytherapy, especially for tumor involving more than 180° around the optic disc.[84] Stereotactic radiotherapy and linear accelerator-based stereotactic RT can target Juxtapapillary uveal melanoma; however, the rate of radiation-related complications is much higher than that of brachytherapy.[88] Smaller tumors can be treated with multiple sessions of TTT to cause tumor shrinkage, but the risk of vision loss, damage to the adjacent retina, and rare extrascleral extension is noted with TTT.[89] Recently, proton beam radiotherapy has been considered as the first-line treatment for Juxtapapillary tumors, which allows delivery of a powerful focused dose of radiation to the tumor when sparing surrounding healthy eye structures.[90]

Follow-up

A long-term follow-up is warranted to look for recurrence and regular examination because Juxtapapillary melanoma can spread via vortex veins to the liver, lung, skin, and brain. Liver and CNS metastases are often associated with optic nerve invasion. Patients with postlaminar optic nerve invasion are at a higher risk of metastasis and melanoma-related mortality.[80,91] The risk of metastasis is marginally higher in patients with postlaminar optic nerve invasion (63%) than in those with prelaminar or laminar optic nerve invasion (53%) or those without optic nerve invasion (61%).[80] Similarly, melanoma-related mortality is higher in patients with postlaminar optic nerve invasion.

Optic disc melanocytoma

Optic nerve head melanocytoma (ONH-MCT), also known as magnocellular nevus of the optic nerve, is a benign melanocytic nevus usually seen in the inferotemporal area of the optic nerve head.[92]

Clinical features

Optic disc melanocytoma is a benign lesion, typically a dark brown-to-black-pigmented tumor located within or adjacent to the optic disc with a feathery margin [Figure 3e]. In total, 15% of tumors are confined to the optic disc, 54% extend to the adjacent choroid, and 30% extend to the sensory retina.[92] Patients with ONH-MCT are usually asymptomatic. Very rarely, patients complain of decrease in vision, which can be secondary to vascular occlusion, malignant transformation, and ischemic tumor necrosis.[92,93] It usually remains stable throughout life or can show progressive growth with malignant transformation seen in 1%–2% of cases.[15,92] When the tumor involves the adjacent retina or choroid, it can mimic other pigmented lesions such as choroidal nevus or melanoma, RPE adenoma, and RPE adenocarcinoma.[94-96]

Spectral-domain optical coherence tomography

SD-OCT of optic disc melanocytoma showed elevated hyper reflective disorganized retinal layers with

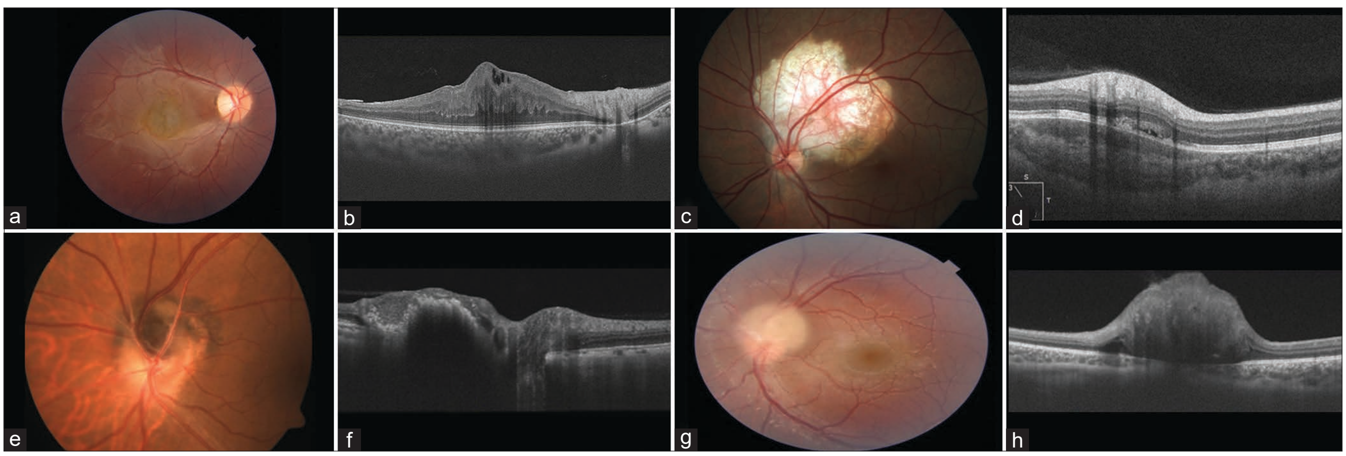

Figure 3: Peripapillary combined hamartoma of the retina and retinal pigment epithelium (RPE), osteoma, melanocytoma, and astrocytic hamartoma. (a) Fundus photograph of the right eye showing a peripapillary greyish lesion with extensive fibrogliotic changes suggestive of combined hamartoma of the retina and RPE (CHRRPE). (b) Optical coherence tomography (OCT) line scan through CHRRPE showing a characteristic inner and outer retinal infolding termed as "omega sign" appearance. (c) Fundus photograph of the left eye showing an amelanotic flat choroidal lesion adjacent to the optic nerve suggestive of peripapillary choroidal osteoma. (d) OCT through the choroidal osteoma showing horizontal lamellar appearance with posterior shadowing. (e) Fundus photograph of the left eye showing a pigmented lesion over the optic disc with feathery margins encroaching into the surrounding choroidal layers suggestive of optic disc melanocytoma. (f) OCT through the melanocytoma showing elevated hyper-reflective lesion with posterior shadowing and overlying disorganized retina. (g) Fundus photograph of the left eye showing an elevated pale-yellow glistening lesion over the optic nerve head with intrinsic calcification suggestive of retinal astrocytoma. (h) OCT through the astrocytic hamartoma showing an elevated tumor-like growth arising from the inner retinal layers with optically empty spaces within the lesion

posterior shadowing with hyperreflective dots within the tumor [Figure 3f]. Microvascular features on OCT-A showed surface vascularization and intrinsic vascularity in the choroidal slab with surrounding hypo reflective boundary.[94] It can differentiate similar-looking pigmented lesions such as nevi or melanoma.

Management

In most cases, treatment is not required. However, it is essential to periodically monitor the tumor because of the possibility of malignant transformation.

Combined hamartoma of the retina and retinal pigment epithelium

Combined hamartoma of the retina and RPE (CHRRPE) is a rare benign tumor, often seen in the Juxtapapillary area. It was first described by Gass in 1973 as a benign pigmented hamartomatous malformation of the retina, RPE, and overlying vitreoretinal interface.[97] They are most commonly seen in males and are almost always unilateral.[98]

Clinical features

CHRRPE can often be seen as an ill-defined gray retinal mass that gradually contracts, which further causes stretching of the blood vessels and retinal striae, ultimately distorting the fovea [Figure 3a].

The spectrum of CHRRPE ranges from vascular, melanocytic, to glial components with epiretinal membrane formation.[99] CHRRPE lesions involving the optic disc or peripapillary area are associated with filigree pigmentary changes, higher grade of retinal distortion or hamartomatous malformation, and CNVM formation than those involving the macula. It is important to distinguish CHRRPE, a benign tumor, from malignant chorioretinal lesions, particularly choroidal melanoma and Rb, as misdiagnoses have led to unnecessary enucleations.[98]

Investigations

Optical coherence tomography

OCT of the peripapillary CHRRPE showed more distortion than that of the macular CHRRPE. CHRRPE can have full-thickness retinal involvement with grossly disorganized retinal morphology [Figure 3b]. OCT can quantify the extent of the outer plexiform layer (OPL), ellipsoid zone (EZ), and RPE involvement with disorganized retinal architecture.[100,101] Pathognomonic OCT signs of CHRRPE include prominent epiretinal membrane, minor vertical vitreoretinal traction (sawtooth configuration/mini peaks), significant vertical vitreoretinal traction with retinal folding of inner retinal layers (maxi peaks) or omega sign, and distortion of the outer plexiform layer.[100-103] The greater extent of RPE involvement could be a potential reason for secondary changes such as CNVM, particularly in patients with Juxtapapillary CHRRPE.[104]

Optical coherence tomography angiography

OCTA of the CHRRPE lesion shows a peculiar mottled, mossy, or filigree vascular pattern. Filigree patterns of small vessels representing abnormal capillaries at the surface of the lesions can be observed clinically and on angiography. A filigree vascular pattern with a high flow is noted in peripapillary CHRRPE lesions with a full-thickness retinal disorganization and minimal preretinal fibrosis.[100,105]

Fluorescein angiography

In the early venous phase, fine blood vessels may be observed within the lesion. These vessels gradually leak fluorescein to stain most of the mass in the late angiograms, although the central portion of the lesion may remain relatively hypofluorescent. Filigree or mosaic pigmentation patterns obscure even larger retinal vessels, leading to hypofluorescence.

Treatment

CHRRPE is a benign tumor, and most will require periodic monitoring. Complete ophthalmologic evaluation is recommended at least every 6 months for patients younger than 12 years. Secondary CNVM-associated CHRRPE can be treated with anti-VEGF injection.[106] Surgery is indicated for the epiretinal membrane and associated vitreomacular traction seen in patients with CHRRPE.[107,108] However, visual recovery is limited due to disorganization of the retinal layers.

Peripapillary choroidal osteoma

Peripapillary choroidal osteoma is a rare, idiopathic, benign peripapillary tumor that primarily affects the choroid. It is characterized by abnormal formation of bone tissue within the choroid [Figure 3c]. It tends to occur more commonly in young adult females.[109]

Clinical features

Most patients remain asymptomatic. However, a few patients may complain of vision loss or distortion, a blind spot in the visual field, or altered color vision secondary to SRF accumulation in the macular area. In most cases, it is a clinical diagnosis, while imaging tests such as OCT and FA might be needed in a few atypical cases [Figure 3d].[110]

Regular follow-up examinations are typically recommended to monitor the size and stability of the tumor and to assess any changes in visual function. Choroidal osteoma located in the peripapillary area can be a developmental tumor. Noble suggested that there may be a congenital, perhaps inherited, abnormality of the peripapillary choroid that is clinically undetectable until calcification begins to occur under the influence of other factors.[111,112]

Treatment

Treatment options for peripapillary choroidal osteoma are limited and mainly depend on the location, size, and symptoms of the tumor. Sometimes, observation and monitoring may be sufficient if the tumor is small and not causing visual disturbances. However, if the tumor is causing significant vision loss or accumulation of SRF in the macular area, treatment options such as laser photocoagulation, PDT, or anti-VEGF injections may be considered.[113,114]

Peripapillary astrocytic hamartoma

Peripapillary astrocytic hamartoma is a benign congenital tumor comprising of fibrillary astrocytes originating from the retinal nerve fiber layer. The incidence is estimated to be 1 in 100,000 individuals without any gender or race predilection. It has been observed in patients as young as 1-year old.[15]

Clinical features

They can appear as yellowish lesions with well-defined borders and small excrescences or as solitary cystic lesions [Figure 3g]. They can be classified into calcified and noncalcified types. The calcified type is characterized by yellow glistening nodules over the optic disc resembling a giant optic nerve drusen. The noncalcified variant is the common type and can mimic early Rb, metastatic tumor, granuloma, or other conditions.[115] Close examination for extraocular signs of tuberous sclerosis complex (TSC) can also facilitate diagnosis. Visual acuity usually remains unaffected, but a blood-filled hamartoma can cause vitreous hemorrhage.

SD-OCT can facilitate the diagnosis and follow-up of peripheral astrocytic hamartoma. OCT characteristic features include a gradual transition from a normal retina into an optically hyperreflective mass with retinal disorganization and posterior shadowing [Figure 3h]. In addition, they can have a "moth-eaten" appearance which are intrinsic optically empty spaces (OES), attributed to intralesional calcification or cavitation.[116,117]

Management

In most cases, observation is needed with periodic follow-up.[118] Secondary CNVM-associated astrocytic hamartomas are treated with anti-VEGF injections.[119] Association of astrocytic hamartoma with a history of seizure will warrant an MRI brain scan to look for associated intracranial hemangiomas. Nonrevolving vitreous hemorrhage may require pars plana vitrectomy.

Leukemias

Leukemias are a heterogeneous clonal disorder characterized by immature myeloid cell proliferation and bone marrow failure.[120] Abnormal myeloid cells can infiltrate the optic nerve more frequently, with acute lymphoblastic leukemia seen in children.[121]

Clinical features

It can be a diagnostic and therapeutic challenge as it can simulate typical or atypical optic neuritis.[122,123] Raised intracranial pressure due to brain involvement of leukemia can mimic optic nerve infiltration. Long-standing cases can have disc pallor. Rarely, optic disc involvement is the initial manifestation of relapse in leukemia [Figure 4d].[122,124] Visual impairment caused by optic disc infiltration can vary depending on the extent of optic nerve involvement.

Management

Optic disc infiltration due to leukemia requires urgent intervention. Diagnostic workup is essential for determining the specific type and staging of the disease. In addition to peripheral smear and bone marrow biopsy tests, flow cytometry, cytogenetic analysis, and molecular testing may also be performed to characterize the disease and guide treatment decisions. The treatment of optic nerve infiltration generally involves prompt systemic and intrathecal chemotherapy, with or without radiation therapy, to the orbits and CNS.[125] However, there is no consensus on the optimal management. In the acute setting, high-dose intravenous and intrathecal methylprednisolone, along with methotrexate and cytarabine, can decrease optic nerve swelling.[126] Newer treatment modalities include immunotherapy with CD3 inhibitors, such as blinatumomab, can promote remission and increase overall survival.[127] The prognosis for individuals with leukemia infiltrating the optic disc depends on various factors, including the type of leukemia, the extent of infiltration, and the response to treatment. Early diagnosis and prompt intervention help improve visual outcomes.[125]

Intraocular lymphoma

Primary intraocular lymphoma (PIOL) is a rare form of non-Hodgkin lymphoma that primarily affects the retina and uvea and rarely involves the optic disc [Figure 4a].[128] Kim *et al*. classified lymphoma involving the optic nerve as (i) primary optic nerve involvement, (ii) optic nerve involvement with CNS disease, (iii) optic nerve involvement with systemic disease, and (iv) optic nerve involvement with PIOL.[128]

Clinical features

Optic disc involvement in lymphoma can be associated with systemic manifestation and concurrent involvement of the brain presenting with focal neurologic

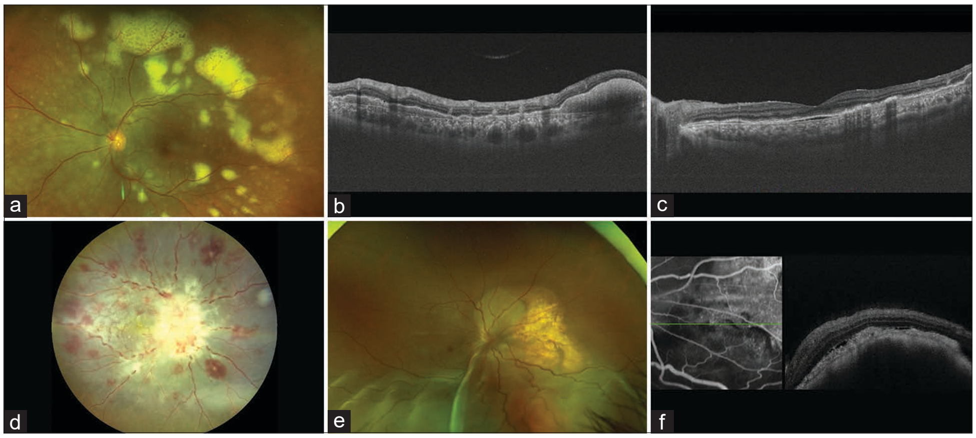

Figure 4: Peripapillary lymphoma, leukemia, and choroidal metastasis. (a) Fundus photograph of a patient with primary vitreoretinal lymphoma with diffuse involvement of the posterior pole showing multifocal, yellow subretinal infiltrates. (b and c) Optical coherence tomography (OCT) showing hyperreflective subretinal infiltration with retinal pigment epithelium (RPE) undulation and sub-RPE deposits. (d) Fundus photograph showing a diffuse pale yellowish lesion infiltrating the optic nerve head along with superficial and deep retinal hemorrhages suggestive of compressive optic neuropathy secondary to leukemia. The patient was detected with chronic myeloid leukemia. (e) Fundus photograph showing a yellowish choroidal lesion resembling leopard-spot appearance surrounding the optic nerve with inferior exudative retinal detachment suggestive of choroidal metastases. The patient was detected with primary lung carcinoma. (f) OCT showing an elevated lumpy-bumpy choroidal lesion with minimal overlying subretinal fluid and thickened RPE

deficits. Increased intracranial pressure due to CNS involvement of lymphoma can mimic optic neuritis.[122,123] Fundus examination can show creamy-white optic nerve infiltration with peripapillary hemorrhages. Long-standing cases can have disc pallor.

Optical coherence tomography

OCT assesses the extent of optic disc involvement and surrounding structures. OCT is an effective tool for detecting subretinal choroidal infiltration in lymphoma. Other OCT characteristics include hyperreflective foci in vitreous, inner, and outer retinal infiltration, undulations of RPE, homogeneous hyperreflective deposits in the sub-RPE space, and pre-RPE nodules [Figure 4b and c].[129,130]

Visual field testing

Visual field testing can detect ischemic optic neuropathy masquerading as optic disc infiltration. Visual field defects can detect the response to lymphoma treatment and monitor its progression.[131] Vertical field defects can also detect associated CNS involvement.

Management

Optic disc infiltration due to lymphoma requires urgent intervention. Diagnostic workups for lymphoma are essential for determining the specific type and stage of the disease. Gadolinium-enhanced fat-saturated MRI can detect the extent of optic nerve and brain involvement. CSF analysis and optic nerve sheath biopsy can reveal false negative results, and when such investigations are inconclusive, biopsy of the optic nerve tissue has a high diagnostic yield. Although rare, lymphomatous optic nerve infiltration must be considered in the differential diagnosis of a pale swollen optic disc.[132] Diagnostic vitrectomy is helpful for cytopathology, flow cytometry, immunohistochemistry, gene rearrangement studies with polymerase chain reaction, and cytokine analysis to confirm the diagnosis of PIOL.[133] Most primary vitreoretinal lymphoma cases involve the CNS; hence, brain MRI and CSF cytology are mandatory. Treatment of optic nerve infiltration generally involves prompt systemic and intrathecal chemotherapy, with or without radiation therapy, to the orbits and CNS.

Metastatic tumors

Metastases to the optic disc are uncommon and account for 4.5% of all intraocular metastases.[134] Metastatic tumors of the optic disc are rare and can have serious implications for vision and overall health. Common primary tumors that can spread to the optic nerve head include breast, lung, and melanoma.[135-139] These tumors can travel through the bloodstream or lymphatic system, eventually reaching the optic nerve head.[134,140]

Clinical features

Metastatic tumors involving the optic disc can compress and damage the optic nerve fibers, leading to visual symptoms such as blurred vision and even total vision loss in severe cases. Less commonly, it can be the initial sign of an underlying cancer without any systemic symptoms.[136,141] Fundus examination shows a swollen optic nerve head and a yellow–white mass overlying the normal structures or a solid mass at the posterior pole with irregular or bumpy contour [Figure 4e]. Metastatic carcinoma may also involve the optic nerve posterior to the lamina cribrosa, where the disc may appear normal, be mildly hyperemic, or reveal slight pallor. This entity of carcinomatous optic neuropathy can mimic retrobulbar neuritis and provide a difficult diagnostic dilemma. In addition, the optic nerve head may be swollen secondary to increased intracranial pressure from metastatic disease to the CNS. Metastatic tumor infiltration of the optic disc can lead to compression of retinal vasculature with retinal venous congestion, increasing the risk of arterial and venous occlusion and subsequent development of neovascular glaucoma. Rarely, metastatic tumors to the optic nerve can also cause significant exudative retinal detachment. The differential diagnosis of optic disc metastases include papilledema, optic neuritis, granuloma of the disc (particularly sarcoid), optic disc drusen, and capillary hemangioma.[134]

Optical coherence tomography

OCT shows a classical description of "lumpy bumpy" surface when the choroid layer is involved [Figure 4f]. It can also detect the extent of disc edema along with adjacent involvement of the choriocapillaris and irregularities of the outer retinal layers.[142]

Fluorescein angiography

Tumor tissue on the disc may appear relatively hypofluorescent in the early phase, whereas in the late phase leakage of dye from the tumor vessels appears as disc hyperfluorescence. A delay in venous filling may be observed if the tumor impairs retinal venous outflow.[15]

Visual field testing

Enlarged blind spots and scotomas can be observed by visual field testing in patients with optic disc metastases. A baseline assessment of the visual field can determine the impact of the tumor on the patient's visual function. Visual field defects can detect the response to treatment and monitor progression.

B-scan

In addition to measuring tumor size and echogenicity, ultrasound can distinguish metastases from other intraocular neoplasms, particularly melanomas. Metastasis involving the optic nerve head can have variable internal structure, which can be regular, irregular, or associated with central excavation on B-scan with moderate to high internal reflectivity on A-scan with no demonstrable internal blood flow.[143]

Neuroimaging

High-resolution chest CT and abdominopelvic CT with contrast play significant roles in the initial screening of primary lesions. It can monitor tumors and can be used for follow-up and surveillance. MRI is a valuable tool for confirming diagnoses, determine the extent of the tumor, and ruling out differentials. It is also helpful in identifying any associated brain metastases. Detailed systemic surveillance is often needed to identify primary tumors elsewhere in the body.

Management

Managing metastatic tumors in the optic nerve head typically involves a multidisciplinary approach involving oncologists, ophthalmologists, and other specialists. Treatment options may include systemic chemotherapy, radiation therapy, targeted therapies, or surgical interventions, depending on the primary tumor type, extent of metastasis, and overall health status.

The prognosis of metastatic tumors involving the optic nerve head can vary depending on various factors, including the type and stage of the primary tumor, response to treatment, and general health. Regular monitoring, close collaboration among medical professionals, and customized treatment plans are crucial to optimize patient outcomes.

When visual acuity is impaired, external beam irradiation (30–36 Gy) delivered over 3–4 weeks may be beneficial in halting the disease progression.

Anterior extension of retrobulbar optic nerve tumors

Optic nerve tumors involving the retrobulbar space seldom involve the optic disc. The most common tumors involving the retrobulbar part of the optic nerve are optic nerve sheath meningioma (ONSM) and optic nerve glioma. ONSM is usually unilateral and is more commonly seen in middle-aged women.[144] On the contrary, optic nerve glioma is most commonly seen in children and adolescents with a possible systemic association with neurofibromatosis type 1.[145]

Clinical features

Patients with retrobulbar optic nerve tumors can present with a transient or painless decrease in vision. Invariably, in 2/3rd of patients, there is a delay in diagnosis either due to inaccurate fundus examination or failure to interpret the neuroimaging findings.[144] Enlargement of tumors can manifest as a decrease in the field of vision, strabismus, and axial proptosis. Fundus examination in the acute stage of disease shows infiltration of the optic disc with peripapillary hemorrhages and disc edema, whereas in the late-stage, the presence of optociliary shunt vessels and optic atrophy is the usual manifestation.[144,146] Rarely, these tumors manifest with retinal vascular occlusion and optic disc neovascularization secondary to compressive and ischemic optic neuropathy.[147]

Neuroimaging

MRI plays an important role in the diagnosis and decision-making of retrobulbar optic nerve tumors. It can reveal the extent of the tumor and is helpful in treatment monitoring and prognosis.[146]

Management

Observation is usually recommended in patients with good visual acuity and smaller lesions. In cases where there is a diagnostic dilemma, biopsy can be performed to confirm the diagnosis. Systemic chemotherapy is usually recommended for small low-grade optic nerve glioma.[145] Different radiation modalities, such as stereotactic radiotherapy, fractionated EBRT, and intensity-modulated radiotherapy, are used to treat ONSM.[148] With current treatment strategies, the overall life prognosis of patients with retrobulbar optic nerve tumors is good with poor visual outcomes.

Conclusion

Tumors either originating or observed close to the optic disc in the peripapillary area pose a diagnostic and therapeutic challenge. Multimodal imaging helps not only document the tumor configuration and vascularity but also differentiate it from simulating optic nerve tumors. As known, optic nerve tumors may be the first manifestations of the underlying systemic disease; hence, as an ophthalmologist, our role is critical not only for diagnosis but also for timely referral to a multispecialist for detailed systemic emulation. With recent advances in medical treatment and the availability of radiation isotopes with focused and targeted delivery of radiation, most tumors can achieve good local control and ocular survival.

Ethical approval

This study was conducted in accordance with the Declaration of Helsinki. The collection and evaluation of all protected patient health information was performed in a Health Insurance Portability and Accountability Act-compliant manner. This study was approved by the Institutional Review board at L V Prasad Eye Institute. Written informed consent was obtained from parent/legal guardian to participate in the study.

Statement of informed consent

Informed consent was obtained from the participants' parent/legal guardian/next of kin to participate in the study prior to performing the procedure, including permission for publication of all photographs and images included herein.

Data availability statement

All data generated or analyzed during this study are included in this article. Further enquiries can be directed to the corresponding author.

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Conflicts of interest

The authors declare that there are no conflicts of interests of this paper.

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